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Innovative Informatics and Artificial Intelligence Research (IIAIR)
Vol. 1, Issue 1, pp. 1–8, 2024
Received 11 Jul 2024; accepted 25 Aug 2024; published 1 Dec 2024
https://doi.org/10.2478/iiair
e-ISSN: XXXX-XXXX

Mangrove Forest Detection in Balikpapan Using YOLOv8 and Google Earth Satellite Imagery

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To cite this article: A. First, A. Second, A. Third, "Title of the manuscript: IIAIR," *Innovative Informatics and Artificial Intelligence Research*, vol. 1, no 1, 2024. [Online]. Available: https://journal.itk.ac.id/index.php/IIAIR

Abstract

Mangrove forests are among the most ecologically important coastal ecosystems due to their role in coastal protection, biodiversity preservation, and carbon sequestration. However, rapid urbanization and land conversion have significantly threatened mangrove ecosystems in Balikpapan Bay, East Kalimantan, particularly in the context of Indonesia's new capital city development. This study proposes a YOLOv8-based object detection approach for automatic mangrove forest detection using very high-resolution (VHR) Google Earth satellite imagery. The proposed framework integrates CLAHE-based image preprocessing with YOLOv8n fine-tuned from COCO-pretrained weights over 250 epochs using the SGD optimizer with cosine annealing scheduling and comprehensive online data augmentation. The dataset comprises 1,314 VHR satellite images partitioned at an 80:20 train-validation ratio, with manual annotations created using LabelImg in YOLO bounding box format. The model performance is evaluated using Precision, Recall, F1-Score, and mean Average Precision (mAP@50 and mAP@50-95). On the validation set, the model attained a precision of 0.687, recall of 0.652, F1-score of 0.669, mAP@50 of 0.584, and mAP@50-95 of 0.309. The results demonstrate the feasibility of YOLOv8-based object detection for high-resolution coastal ecosystem monitoring and are expected to support sustainable environmental management in Balikpapan.

Keywords: machine learning; YOLOv8; mangrove detection; satellite imagery; remote sensing; object detection

Introduction

One of the most productive and ecologically valuable coastal areas in tropical regions is mangrove forests that provide ecological services including carbon storage, coastal erosion protection, and marine biodiversity habitat [2]. However, despite the ecological and socio-economic significance of these forests, there have been many losses in mangrove area in recent years caused by the growth of aquaculture, urbanization, and land transformation [5]. One of the regions that needs special attention is the Balikpapan Bay coastal region in the province of East Kalimantan due to the presence of large mangrove ecosystems and development challenges connected to the construction of a new capital of Indonesia, IKN Nusantara, in neighboring regency, Penajam Paser Utara.

Remote sensing techniques are currently used as the primary methods for mangrove mapping but the traditional methodologies using thresholding of spectral index along with the use of medium resolution data such as Landsat and Sentinel are inadequate in capturing individual patches and their boundaries [6, 2]. Estimation of mangroves' above-ground carbon storage in the Indonesian small islands is another study conducted using vegetation indices calculated from Sentinel-2 images, which is an indicator of reliance on the medium resolution platforms [20]. Very High Resolution (VHR) images such as those available from Google Earth can play a crucial role in improving detection accuracy by enabling recognition of individual mangrove crowns and their structural details [9].

Deep learning techniques have revolutionized remote sensing image interpretation through automated feature extraction without spectral index definitions, with the use of encoder-decoder models like U-Net being common for mangrove mapping at the pixel level [11, 10] with recent work on using UNet++ for mangrove dynamics estimation from multi-temporal Sentinel-2 images [19]. Yet, due to their high computational complexity over a broad range of geographic locations, there is a need to implement more computationally efficient object detection systems. YOLO object detection algorithms provide an efficient way of predicting bounding boxes in one step that allows faster inference for operational use [13]. YOLOv8 algorithm, published in 2023 by Ultralytics, has an anchor-free detection head and improved backbone model structure [13].

Despite demonstrated YOLO performance in vegetation and land cover detection from aerial imagery [8, 4, 12], including UAV-based detection of mangrove species using YOLOv5 in Sarawak, Malaysia [14], no published study has specifically applied YOLO to mangrove detection using VHR Google Earth imagery in Balikpapan. Therefore, this study proposes a YOLOv8-based framework for automatic mangrove forest detection using Google Earth satellite imagery of Balikpapan Bay. The main contributions of this study are as follows:

- A VHR satellite image dataset of 1,314 Google Earth images of Balikpapan Bay annotated with mangrove

bounding boxes in YOLO format.

- A preprocessing pipeline integrating CLAHE contrast enhancement with YOLOv8n fine-tuning from COCO-pretrained weights.
- A quantitative evaluation of the YOLOv8n model on bounding-box detection metrics (Precision, Recall, F1-Score, mAP@50, mAP@50-95) on a held-out validation set.

Existing mangrove monitoring studies in Indonesia and Kalimantan primarily rely on medium-resolution satellite imagery such as Sentinel-2 and Landsat. While these approaches are effective for large-scale mapping, they are insufficient for detecting fine-scale mangrove boundaries and individual vegetation structures. Moreover, previous studies have focused mainly on semantic segmentation and pixel-based classification approaches rather than object detection methods, and existing YOLO-based mangrove detection work has so far been limited to UAV-acquired imagery [14] rather than satellite-derived VHR sources. To the best of our knowledge, no published study has specifically applied YOLOv8-based object detection for mangrove forest detection using Google Earth very high-resolution imagery in Balikpapan Bay. This study addresses this gap by proposing an automated YOLOv8 framework for mangrove detection using high-resolution satellite imagery.

Related Works

Mangrove detection from satellite imagery sits at the intersection of ecological remote sensing, deep learning, and object detection research. The following subsections review prior contributions in each of these domains, tracing the progression from traditional spectral methods to modern single-stage detectors, and identifying the specific gap that motivates this study.

Mangrove Monitoring with Remote Sensing and Machine Learning

Remote sensing has become the primary tool for large-scale mangrove monitoring due to its cost efficiency and extensive spatial coverage. Threshold segmentation methods based on vegetation indices have demonstrated good performance for mangrove extraction, although their effectiveness strongly depends on local threshold calibration [2]. Machine learning techniques have further improved mangrove mapping performance. Ghorbanian et al. [6] combined Sentinel-1 and Sentinel-2 imagery with Random Forest classification in Google Earth Engine to generate accurate mangrove ecosystem maps at national scale. In Indonesia, Al Hakim et al. [5] applied machine learning approaches using Sentinel-2 imagery for mangrove forest change mapping in Balikpapan Bay, directly motivating the present study's focus on the same geographic region. Sudiana et al. [7] demonstrated that SAR and optical data fusion with deep learning can significantly improve mangrove mapping accuracy. Karang et al. [20] further showed that Random Forest classification combined with Sentinel-2 vegetation indices can effectively map and estimate mangrove above-ground carbon stocks on small Indonesian islands, illustrating the continued relevance of machine learning approaches for blue carbon assessment.

Deep Learning for Mangrove Segmentation

Deep learning approaches have substantially advanced remote sensing analysis through automated feature extraction. Win and Sasaki [11] applied U-Net for mangrove change detection using medium-resolution satellite imagery and achieved high classification accuracy. Fu et al. [10] compared SVM and Res-UNet approaches for spatiotemporal mangrove analysis and showed that deep learning models provide more spatially consistent delineation results. Seydi et al. [3] proposed a hybrid convolutional neural network for land cover mapping in mangrove ecosystems using multi-temporal Sentinel-2 imagery. A similar UNet++ based monitoring system applied to Sentinel-2 time series in the United Arab Emirates demonstrated the capability of deep segmentation networks to track multi-year mangrove expansion and associated carbon sequestration gains [19]. Studies using very high-resolution imagery have also shown improved capability for identifying mangrove boundaries and vegetation structures [9]. While these segmentation-based approaches achieve high pixel-level accuracy, their computational overhead over large spatial extents motivates adoption of single-stage object detection frameworks such as YOLO.

YOLO-Based Object Detection in Remote Sensing

YOLO model-based object detection is performed via one single feed-forward pass that provides fast and efficient inference capability compared to other multi-stage object detectors [8]. In addition, the new generation YOLOv8 detector from Ultralytics released in 2023 achieves better speed and higher detection accuracy due to anchor-free detection strategy and optimized backbone architecture [13]. The usage of YOLO models for remote sensing tasks provided promising results for analysis of aerial and satellite images. For example, Li et al. [8] used the improved attention mechanism for RSI-YOLO model for remote sensing object detection task. Vaghela et al. [4] used YOLOv8 detector for agricultural land cover classification with promising detection accuracy. Topgül and Sertel [12] evaluated YOLO-based models on the VHRTrees dataset for high-resolution tree detection and presented strong mAP result for vegetation object detection. Concerning YOLO model application for mangrove ecosystem, Lim et al. [14] detected dominant *Rhizophora* mangrove crowns using YOLOv5 from consumer UAV-based RGB imagery in Sarawak, Malaysia, providing the true positive rate above 70%.

Beyond vegetation monitoring, Zhang et al. [15] proposed MD-YOLOv8, which integrates a multi-frequency attention downsampling module into the YOLOv8 backbone to improve multi-object detection accuracy on the DOTA satellite imagery benchmark, while Chi et al. [16] introduced EP-YOLOv8 with a large selective kernel block for detecting electric pylons in optical satellite imagery, both illustrating the broader trend of customizing YOLOv8 backbones for satellite-scale targets. YOLOv8 has also been reported to outperform other YOLO variants such as YOLOv7 and YOLOv12 for small object detection in synthetic aperture radar (SAR) satellite imagery, highlighting its robustness across imaging modalities [17]. In a forestry context closely related to vegetation health monitoring, Shi et al. [18] proposed YOLOv8-MFD, which fuses a MobileViT-based backbone with a Focal Modulation mechanism to detect pine wilt diseased trees from UAV

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imagery, reporting a mAP@0.5 of 88.2%.

Methods

The proposed methodology for the research involves the following steps: (1) identifying the study area and data collection; (2) data preparation and annotation; (3) image preprocessing; (4) design and evaluation of the model; and (5) model testing and evaluation.

YOLOv8 Architecture

YOLOv8, launched by Ultralytics in 2023, is an anchor-free object detector that makes bounding box predictions in a single forward pass [13]. The backbone network uses a C2f (Cross Stage Partial with Two Bottleneck Connections) module to extract features at multiple scales efficiently. These features are subsequently fused using a Path Aggregation Network (PAN) neck before feeding into decoupled heads responsible for predicting object classes and bounding box offsets without utilizing any anchor boxes [13]. The YOLOv8 variant chosen for this paper among various options was the Nano (YOLOv8n) due to its high computational efficiency, allowing efficient execution of the model on hardware-constrained environments, providing good detection performance in single-class satellite imagery [4, 12].

Study Area

The current study concentrates on the coastal zone of Balikpapan Bay, located in East Kalimantan, Indonesia, and which falls under the geographical coordinates of 1.0°–1.5°South latitude and 116.7°–117.0°East longitude. The Balikpapan Bay is home to diverse coastal ecosystem zones such as the mangroves, aquacultures, tidal zones, and the outskirts of the city. The climatic conditions in the area are characterized by high humidity and rainfalls; it has large tides that influence the location and structure of the mangroves [9].

Dataset

High resolution satellite images (VHR) were collected through an organized approach of screenshotting using Google Earth Pro in coastal areas of Balikpapan Bay where mangrove forest ecosystems are known to exist. The collection of the data entailed collecting clear images, where the images had a spatial resolution that was in the range of 0.5 m to 1.0 m per pixel. The spatial resolution is similar to the spatial resolution of images used in classifying VHR as resolving individual mangrove canopy structures [9, 1]. A total of 1,314 images were collected independently by the project team.

Annotations were performed manually via the LabelImg annotation software that employs the YOLO format for annotating objects in terms of normalized bounding boxes based on their coordinates (*x_center*, *y_center*) and size, which is expressed in terms of the width and height in relation to total image sizes [13]. Each individual bounding box was employed to annotate only one single mangrove cluster in the image by attaching one single object label (*mangrove*). Manual annotation was performed by project members with domain expertise in mangrove habitats to avoid mislabeling spectrally similar vegetation species [12], the same being stressed also in case of unmanned aerial vehicle-based mangrove annotation [14].

The image database consisting of 1,314 images was then divided into training and validation datasets at a ratio of 80:20 using the split-folders library with a fixed random seed of 42 to ensure reproducibility, leading to a training set of 1,051 images and a validation set of 263 images. The split ratio conforms to common practices in remote sensing deep learning applications as indicated in [1, 5].

Image Preprocessing

Before model training, all images underwent a preprocessing pipeline tailored to the characteristics of VHR satellite imagery. Images were first resized to a uniform size of 960×960 pixels, consistent with the input dimension configured for this study [13]. To enhance image contrast and improve visual distinction between dense mangrove canopies and surrounding water or land features, Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied to the luminance (L) channel of each image in the LAB color space, with a clip limit of 2.5 and a tile grid size of 6×6. Pixel intensities were subsequently normalized to the range $[0, 1]$ following the standard YOLOv8 input pipeline.

In the course of training, online augmentation of data using the data augmentation pipeline present in the Ultralytics YOLOv8 architecture was used to increase the generality and robustness of the model towards input data variations [8, 4]. Data augmentation methods that were used include: horizontal and vertical flipping (probability 0.5 each); random rotation of up to $\pm 5°$, chosen conservatively to avoid excessive distortion of canopy texture; HSV color-space jittering with hue, saturation, and brightness adjustment factors of 0.01, 0.5, and 0.3, respectively; random scaling ($\pm 30\%$) and translation ($\pm 10\%$); mosaic augmentation (probability 0.8, disabled during the final 20 epochs to stabilize convergence); mixup augmentation (probability 0.1); and copy-paste augmentation (probability 0.1).

Experimental Settings

The YOLOv8n model was pre-trained with COCO weights (yolov8n.pt) in order to benefit from strong low-level features that can be transferred to the satellite images [4, 12]. All experiments were performed in Google Colab with one GPU (device=0). The dataset splitting process was done with a fixed seed of 42 by the split-folders library in order to fully reproduce train/validation splitting procedure.

The model fine-tuning was performed for 250 epochs with batch size of 8 and image size of 960×960. The SGD optimizer was used with an initial learning rate of $lr_0 = 0.01$, final learning rate factor of $lrf = 0.01$ (resulting in a cosine-annealed terminal learning rate of $lr_0 \times lrf = 1 \times 10^{-4}$), momentum of 0.937, and weight decay of 5×10^{-4}. Cosine annealing with a warm-up of 5 epochs (warm-up momentum = 0.8) was used as a learning rate schedule. The loss weight for the bounding box localization and classification was set to $box = 9.0$ and $cls = 0.4$, respectively, to put more importance on the first task in relation to classification, which is reasonable for the single-class object detection.

The checkpoint with the highest performance was chosen by the Ultralytics model training framework using the highest mAP@50 on validation set obtained from all epochs and was saved as best.pt. The final evaluation was performed using the function model.val() on the best checkpoint with confidence threshold at 0.25 and IoU threshold of 0.6 on the validation set.

Table 1: Key training hyperparameters for the YOLOv8n mangrove detection model

Hyperparameter	Value
Base model	YOLOv8n (COCO-pretrained)
Epochs	250
Image size	960×960
Batch size	8
Optimizer	SGD
Initial learning rate (lr_0)	0.01
Final LR factor (lrf)	0.01
Momentum	0.937
Warm-up momentum	0.8
Weight decay	5×10^{-4}
Box loss weight	9.0
Class loss weight	0.4
LR scheduler	Cosine annealing
Warm-up epochs	5
Early stopping patience	50
Random seed (dataset split)	42
Confidence threshold (eval)	0.25
IoU threshold (eval)	0.6
Model selection criterion	Best mAP@50 on validation set

Performance Evaluation

The following metrics were used for the evaluation of model accuracy: Precision (P), Recall (R), F1-score, and mAP. Precision is the ratio of true mangrove object detections to the total number of positive predictions made by the model, which can be formulated as Equation 1, where TP stands for the number of true detections, and FP – false positives. Recall is the percentage of ground-truth mangrove objects that have been detected by the model, as indicated by Equation 2, where FN means the number of mangrove objects not detected by the model.

$$Precision = \frac{TP}{TP + FP} \tag{1}$$

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

The F1-Score was computed as the harmonic mean of Precision and Recall to provide a single balanced measure of detection performance, as shown in Equation 3.

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{3}$$

Mean Average Precision (mAP) was calculated at an IoU threshold of 0.50 (mAP@50), as shown in Equation 4, and across multiple IoU thresholds from 0.50 to 0.95 (mAP@50-95), as shown in Equation 5, to assess detection performance under varying localization criteria [8, 12]. An IoU threshold of 0.50 was used as the minimum localization requirement for a true positive detection.

$$mAP@50 = \frac{1}{C} \sum_{c=1}^{C} AP_c \Big|_{IoU=0.50} \tag{4}$$

$$mAP@50-95 = \frac{1}{10} \sum_{t \in \{0.50, 0.55, \ldots, 0.95\}} mAP_{IoU=t} \tag{5}$$

Results and Discussion

This section presents the quantitative detection results of the YOLOv8n model, followed by qualitative analysis of detection outputs and training convergence behavior.

Detection Performance

Table 2 summarizes the validation results of a trained YOLOv8n model. This gives us a Precision of 0.687, Recall of 0.652 and F1-Score of 0.669 for the model. This model achieved a mean Average Precision at IoU 0.50 (mAP@50) of 0.584 for localization, however the more stringent mAP@50-95 averaged across the IoU thresholds from 0.50 to 0.95 was much lower, reaching only 0.309 as shown in Figure 1.

Table 2: YOLOv8n mangrove detection evaluation metrics on the validation set

Metric	Score
Precision	**0.687**
Recall	0.652
F1-Score	0.669
mAP@50	0.584
mAP@50-95	0.309

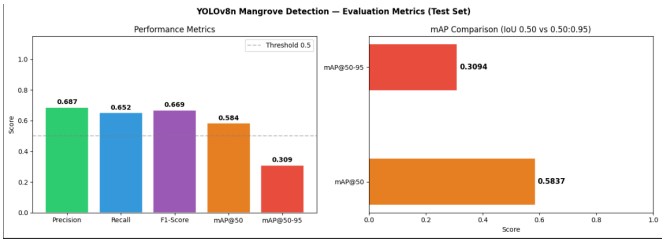


Figure 1: YOLOv8n mangrove detection evaluation metrics on the validation set: (left) Precision, Recall, F1-Score, mAP@50, and mAP@50-95; (right) comparison between mAP@50 and mAP@50-95.

Qualitative Detection Results

Qualitative evaluation was done through a comparison between the ground truth annotations and the YOLOv8n detections on validation images that have been presented in Figure 2. The left side illustrates ground truth bounding box annotations (in green) whereas the right side illustrates detected bounding boxes (in blue).

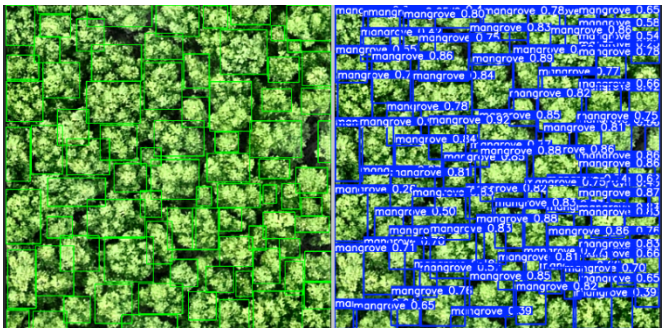


Figure 2: Qualitative comparison between ground-truth annotations (left, green) and YOLOv8n predicted detections with confidence scores (right, blue) on a representative densely vegetated mangrove tile.

Even with the dense vegetation cover in the validation tile, the algorithm performs well in localizing the patches of the

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mangrove canopy with bounding boxes that mostly match with ground-truth annotations. The detection scores differ greatly between 0.14 and 0.92, showing that the algorithm gives high scores when the crowns are identifiable and low scores when there is dense vegetation cover in the canopy. Some ground-truth annotations lack the predicted box by the model, whereas some predicted boxes intersect with the surrounding crowns.

Training Convergence and Detection in Densely Clustered Canopy

The plot in Figure 3 shows the evolution of loss during the training and validation processes for the YOLOv8n architecture after 250 epochs, which include box loss, classification loss, and distribution focal loss (DFL).

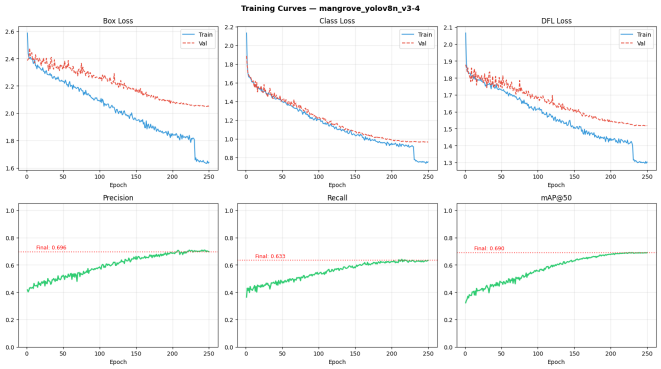


Figure 3: Training curves for YOLOv8n mangrove detector on 250 epochs for box loss, class loss, and DFL loss for training set and validation set (first row), validation Precision, Recall, and mAP@50 performance (second row).

All three loss metrics exhibit a consistent decline during training, where the cosine annealing learning rate results in a distinct decay graph that allows the model to converge more smoothly in the later epochs. A small difference between training and validation loss graphs, especially the box loss one, indicates overfitting, which is caused by the comparatively small size of the training dataset used for VHR Google Earth. The Precision, Recall, and mAP@50 validation metrics continue to improve until about epoch 200.

This convergence behavior and model accuracy can be further understood by analyzing the most challenging detection scenarios. Figure 4 depicts YOLOv8n predictions on an extremely dense mangrove tile where individual tree crowns are visually contiguous.

On this dense tile, the model generates bounding boxes with confidence scores ranging from 0.30 to 0.88. The high degree of box overlap and label crowding reflects the core difficulty of detecting mangrove patches that are visually contiguous: the ambiguous canopy boundary produces multiple candidate bounding boxes for the same physical cluster. Relaxing the NMS (Non-Maximum Suppression) threshold could partially mitigate excessive overlap, while architecture-level solutions such as multi-scale attention mechanisms [15, 16] represent more principled future directions.

Discussion

Since the relatively close Precision and Recall (0.687 and 0.652, correspondingly), there is no bias towards mangrove detection or the omission of false positive cases. Therefore, the F1-Score of 0.669 is indicative of an average detection quality of the model considering that the size of the VHR

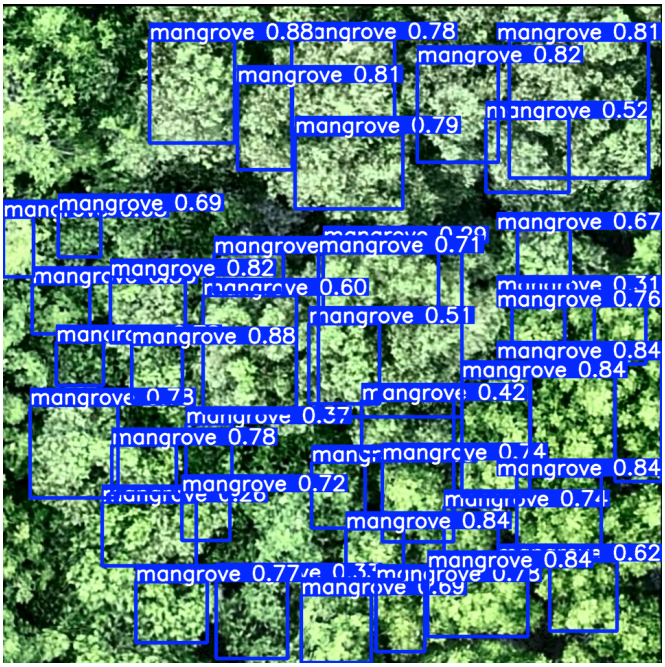


Figure 4: YOLOv8n predicted bounding boxes with confidence scores on an extremely densely clustered mangrove canopy tile, illustrating substantial box overlap and label crowding.

training data set is relatively small in comparison with the large data sets used as a benchmark by Zhang et al. [15] and Karwowska et al. [17].

The difference between the mAP@50 score (0.584) and mAP@50-95 score (0.309) is the most relevant diagnosis of the problem. Though the model can easily localize the patches of mangroves at a low Intersection-over-Union score, its ability to generate accurate bounding boxes deteriorates significantly when a strict localization threshold is set. Such a feature is understandable given the nature of the target – the irregular, continuous boundary of mangroves, whereas more separate and geometrically precise objects, such as electric pylons [16] or diseased pines [18], have significantly higher mAP@50-95 score in YOLO-based detectors. Similar problems of bounding boxes accuracy in detecting mangrove crowns from aerial photos using YOLOv5 were encountered by Lim et al. [14]. The authors reported high true positive rates above 70 percent but still low accuracy of bounding boxes.

These results altogether prove that the proposed YOLOv8n model is a feasible starting point for mangrove object detection from satellite images in Balikpapan Bay. The difference in mAP@50-95 values can be an encouragement for a number of improvements, including enlarging the dataset of annotations, using multispectral images in order to get more spectral differences between the mangroves and other plants, developing guidelines for ambiguous canopy boundaries, as well as changing the backbone similar to MD-YOLOv8 [15] and EP-YOLOv8 [16].

Conclusion

This study proposed and evaluated a YOLOv8-based framework for automated mangrove forest detection using VHR Google Earth satellite imagery of Balikpapan Bay. The fine-tuned YOLOv8n model, trained on a custom dataset of 1,314 annotated satellite images with CLAHE preprocessing and

comprehensive online augmentation, attained a Precision of 0.687, Recall of 0.652, F1-Score of 0.669, mAP@50 of 0.584, and mAP@50-95 of 0.309 on the validation set.

The findings support the use of one-step object detection for very high resolution observation of mangroves in coastal regions, and the substantial difference between mAP@50 and mAP@50-95 shows that proper identification of the canopy edge is still one of the major problems. Possible future work should include enlarging the training set by increasing its geographic diversity, incorporating multispectral or SAR images to enhance the spectral discrimination power, improving the process of labeling ambiguous canopy edges, and comparing the performance of YOLOv8n with other models such as YOLOv8-seg.

Acknowledgments

The authors would like to thank Institut Teknologi Kalimantan for supporting this research.

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